

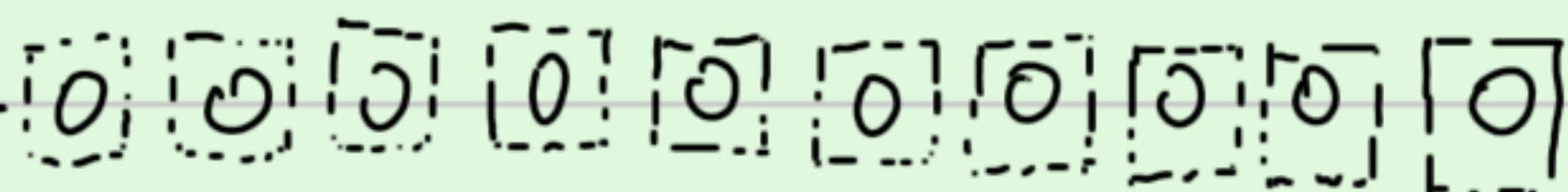
back in time to ancestral generations



parent



tossing $n=10$ balls $\uparrow$ randomly into i (randomly select from $2N$) boxes, $i=10, 9, \dots$



$n=10 \ll 2N$

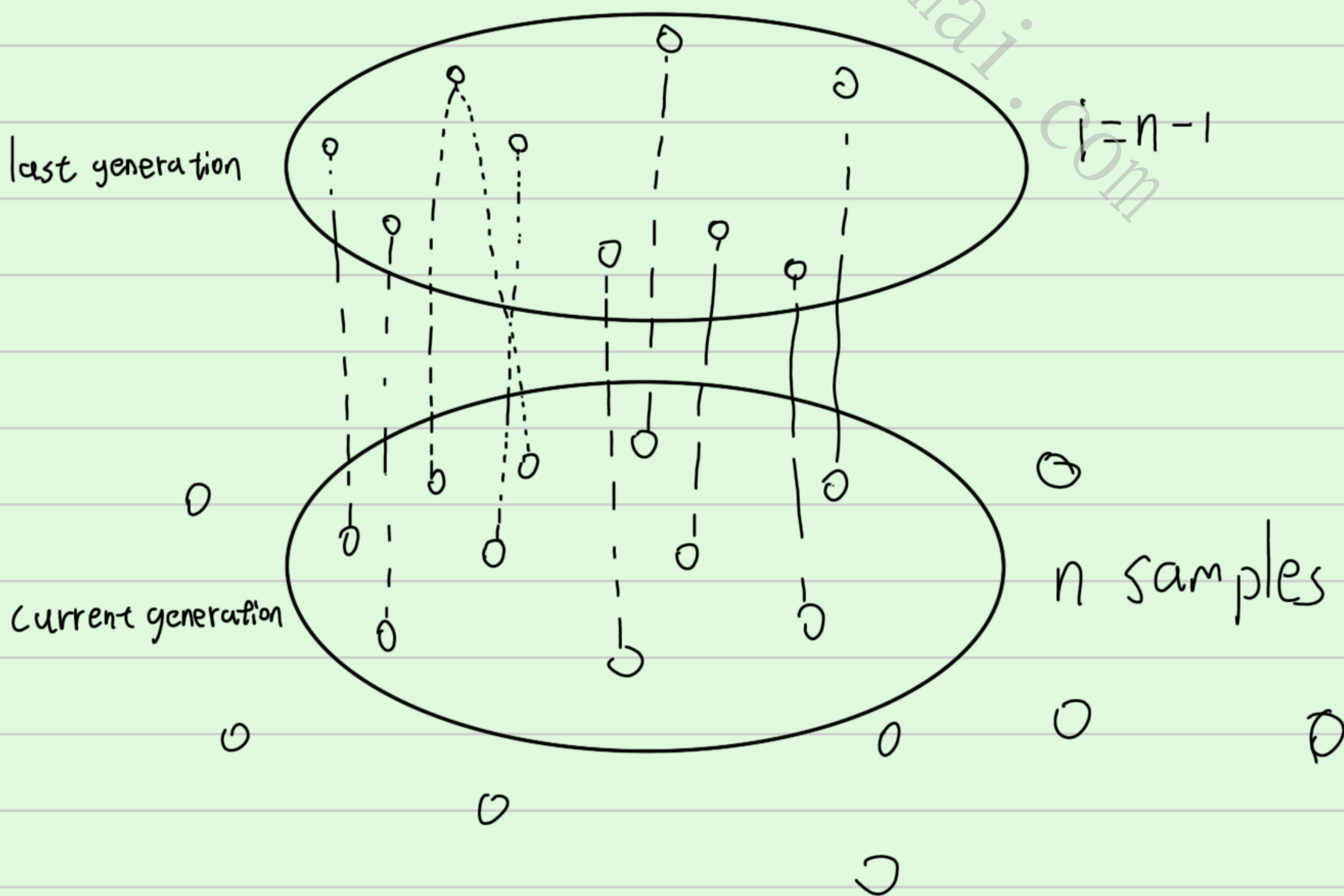
current generation

$$G_{n,i} = \frac{\sum_n^{(i)} N_{[i]}}{N^n}$$

$$\left\{ \begin{array}{l} G_{n,n} = 1 - \frac{C_n^2}{2N} + o\left(\frac{1}{N}\right), \quad i=n \\ G_{n,n-1} = \frac{C_n^2}{2N} + o\left(\frac{1}{N}\right), \quad i=n-1 \end{array} \right.$$

$$N \rightarrow \infty \Rightarrow G_{n,n} + G_{n,n-1} = 1$$

N generation as unit : $G_{n,n-1} \rightarrow$



1. 溯祖事件

当 $N \rightarrow \infty$, 每代溯祖 ($G_{n,n-1}$) 或不溯祖 ($G_{n,n}$) 视作 Bernoulli trials

X	coalescent	not
p	$1 - \frac{C_n^2}{2N}$	$\frac{C_n^2}{2N}$

(单位时间为 N)

t 代发生 k 次的概率

$$P_t \{k=k\} = C_t^k (G_{n,n-1})^k (G_{n,n})^{t-k} \xrightarrow{\lambda = t G_{n,n-1}} \frac{\lambda^k}{k!} e^{-\lambda}$$

coalescent 事件的等待时间

$$P\{T=t\} = G_{n,n-1} (1 - G_{n,n-1})^t \xrightarrow[\text{概率为 } \lambda \Delta t]{\Delta t \text{ 内发生}} \lambda \Delta t (1 - \lambda \Delta t)^{t/\Delta t}$$

变为

根据 $x=0$ 处 e^{-x} 的 Taylor 展开, 当 x 很小时 $e^{-x} \approx 1 - x$

得到 n 个样本溯祖一次的等待时间 T_n 的密度函数为

$$\Delta t \lambda e^{-\lambda \Delta t \cdot t/\Delta t} \xrightarrow{\Delta t \lambda e^{-\lambda t}} f_{T_n}(t) dt = \lambda e^{-\lambda t} dt \quad (\lambda = \frac{C_n^2}{2N})$$

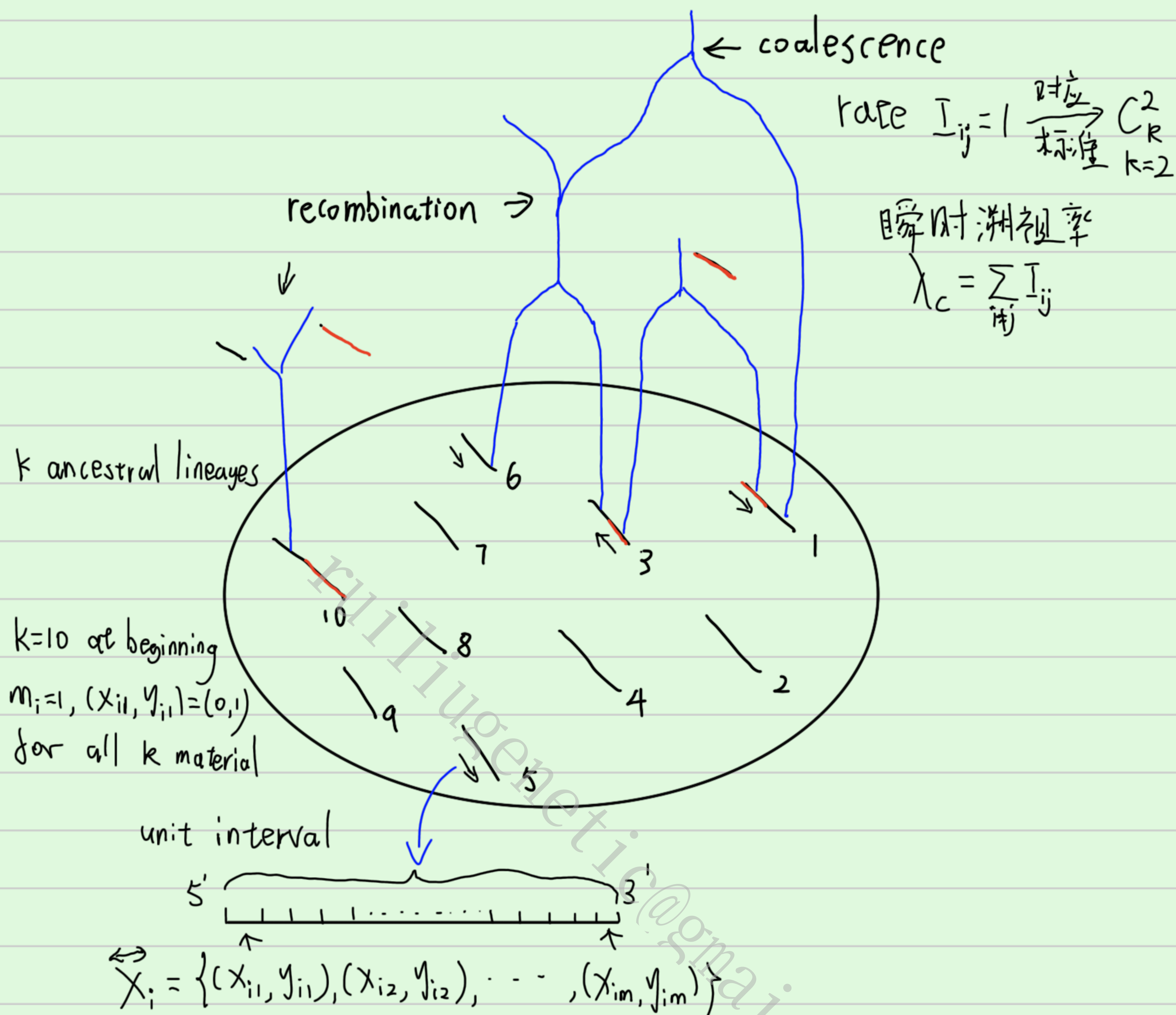
以 $2N$ 代为单位时间, $\lambda = C_n^2$

$$E[T_n] = \frac{1}{\lambda} = \frac{2}{n(n-1)}, \quad \text{Var}[T_n] = \frac{1}{\lambda^2}$$

n 个样本的总溯祖时间 (ie. 树高) $T_{\text{MRCA}} = T_n + T_{n-1} + \dots + T_2$. 且各次溯祖事件相互独立

$$E[T_{\text{MRCA}}] = \sum_{i=2}^n E[T_i] = \frac{2}{n(n-1)} + \frac{2}{(n-1)(n-2)} + \dots = 2(1 - \frac{1}{n}) \quad \text{for both W.F. \& Moran model}$$

$$E[T_{\text{total}}] = 2 \sum_{i=2}^n i \frac{2}{i(i-1)}$$



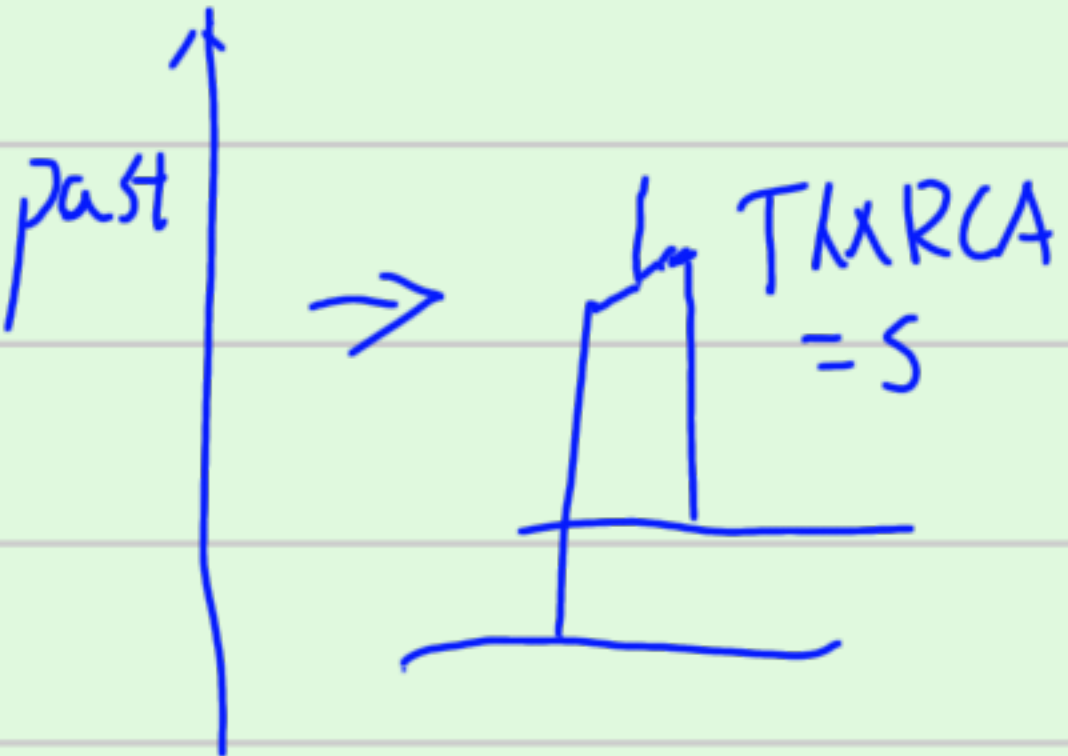
after coalescence, if any interval remain just one ancestral lineage i.e. only the MRCA of that interval, $k=1$ for the interval, remove the interval.

(若某区间在溯祖之后, 所有样本合为一个MRCA, 则移除该区间)

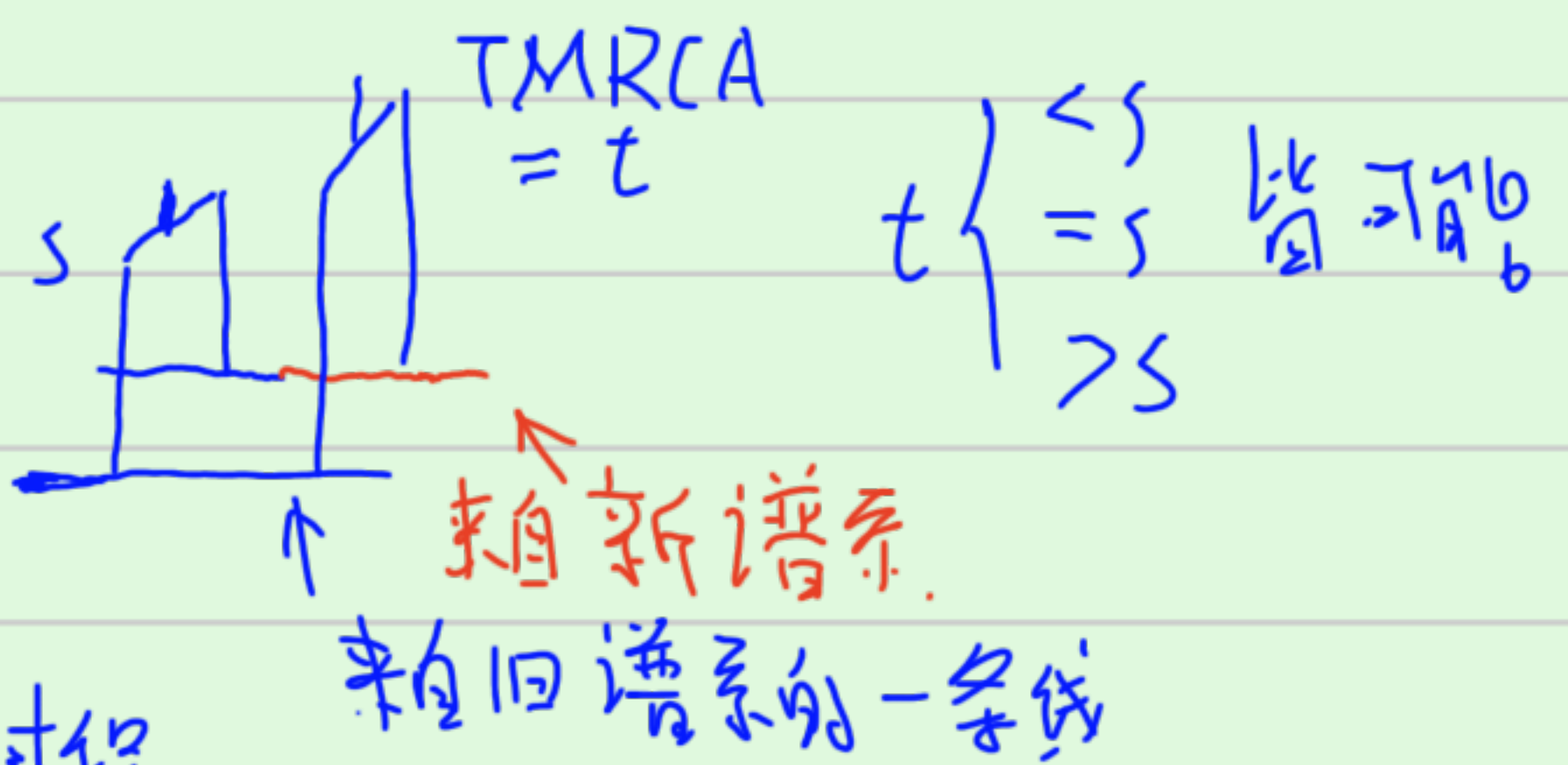
停止 无共祖片段之间溯祖. 即 $I_{ij}=0$, 如果两片段间无重叠 ancestral 片段

PSMC 将染色体从左至右的每个 bin 的 TMRCA 作为 HMM 隐状态.

bin 中是否含有杂合子为观测值. 0, 1.



在 $(0, s)$ 内发生重组
则下一个 t 发生状态转移



新谱系线与旧线的共祖时间按照 coalescent 过程
从现在往过去重算

$$f(t) = \lambda(t) e^{-\int_0^t \lambda(u) du}$$

PSMC 中 $\frac{1}{5}$ 倍

